

UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS CIVITAS 1.0

Living City Graph, Infrastructure, Accessibility and Verified Change
Cartridge

UGTOMS CIVITAS 1.0 Architecture

A city is a layered graph of places, routes, utilities, rules, observations and verified change.



One city can hold observed, normative and simulated layers without collapsing them into one truth.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

ugtoms.domain.civitas@1.0.0

Component	CIVITAS
Canonical identity	ugtoms.domain.civitas@1.0.0
Component type	Domain cartridge
Version / date	1.0.0 / 30 August 2026
Document type	Open modular cartridge specification and implementation-ready profile
Status	Implementation-ready contract with synthetic deterministic fixtures; no municipal, accessibility, emergency or regulatory deployment approval claimed

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

Section	Purpose
1	Executive decision and cartridge promise
2	Source basis and evidence discipline
3	Open modular cartridge ABI
4	Formal city object and authority classes
5	Stable place, asset and actor identity
6	Mobility, accessibility and route topology
7	Utilities, capacity and infrastructure dependencies
8	Policies, permits, schedules and stewardship
9	Observation, simulation and normative authority split
10	Verified city-event and atomic-patch pipeline
11	Deterministic festival-outage-flood demonstration
12	Queries, packing, privacy and external adapters
13	Reference validation, boundaries and kill criteria
14	Complete 48-mechanism catalog
15	Claims ledger
16	High-impact profiles and final definition
READING RULE	
The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.	

Executive Decision

DECISION - GO
Release CIVITAS as the first Batch 03 cartridge. Make the city itself a typed, replayable world object: streets, rooms, utilities, schedules, accessibility, ownership, observations and scenario branches may interact, but only verified city patches alter authoritative state.

The dope part is not another flat digital twin. CIVITAS treats a city as a playable, inspectable rule-world whose infrastructure, permissions, routes and history are first-class objects. A player, planner, scanner, simulation or sensor can propose a radical change; none may silently rewrite the city.

Release promise	CIVITAS 1.0 result
One layered city graph	Places, routes, utilities, policies, observations and scenario branches share stable references without losing type or authority class.
Accessibility is a query dimension	Routes are evaluated under explicit movement, clearance, surface, lift, sensory, communication, effort, schedule and capacity profiles.

Release promise	CIVITAS 1.0 result
Infrastructure is causal only by declaration	Power, water, transit and service impacts follow an explicit dependency graph, not proximity or visual overlap.
Observation is proposal-only	Scans, inspections, models and human reports retain uncertainty and provenance until a verifier promotes a bounded patch.
Simulation cannot impersonate reality	Construction, festival, disaster and future-plan branches remain visibly simulated unless separately authorized and evidenced.
Every accepted change is replayable	Atomic patches carry reason codes, pre/post hashes, conflicts, evidence and lineage.

Source Basis and Evidence Discipline

The cartridge inherits a common UGTOMS rule: domain knowledge may be broad, but every authoritative change stays typed, ordered, evidence-bounded and replayable.

- Definitions, operators and profiles live inside the substrate with stable IDs, explicit dependencies and content hashes [S1-S3].
- Pure evaluation and model output remain upstream of support, compatibility, guards, verified proposals, deterministic commit and lineage [S1, S2, S4-S7].
- Coordinates, glyphs, pictures, meshes, simulations, scans and packed keys are not identity or authority by themselves [S1, S4-S10].
- Exactness, uncertainty, capability, budgets and error margins are explicit; UNKNOWN and REJECTED are valid outcomes [S1, S2, S5, S8-S10].
- Component-scoped identities keep this cartridge from masquerading as the newest universal version of every UGTOMS track [S11].
- Existing specialist cartridges remain peers and adapters; this report does not silently absorb their schemas or claims [S13-S17].

ENGINEERING-DERIVED DOMAIN CONTENT

The municipal, infrastructure, mobility, schedule, accessibility and policy records in CIVITAS are new engineering definitions. The source corpus contributes stable identity, topology, uncertainty, evidence promotion, deterministic commit, replay, typed operators and component-scoped governance; it does not claim a finished city platform.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS__FOUNDATION__UNICODE_v5.pdf	Foundation 5.0 OperatorCells: literal, glyph, semantic, field and converse faces remain co-addressed but typed separately.	pp. 1-21
S2	UGTS_KC_4_2_GeneralOperator_Order_Addendum(1).pdf	Typed OperatorSpec, dependency/phase/event/replay order, exactness, capability, budgets and adapter fallback policy.	pp. 2-32
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as content-addressed substrate data, resolvable references, dependency closure and explicit operation order.	pp. 1-15
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → verified event → transition → lineage authority and packed-state boundaries.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Patterns, fields, topology, SE(2)/SE(3), bounded dynamics, hybrid events and numerical/error contracts.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable scene identity, proposals, deterministic commit, checkpoints, rollback, replay and divergence detection.	pp. 3-19
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic fixed-step runtime, snapshots/hashes and self-contained browser delivery.	pp. 2-16
S8	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Time/phase/winding, finite topology maps, interval guards, finite grammar and exact packed-key boundaries.	pp. 1-15
S9	UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf	Spatial observation versus authority, stable nodes/edges, uncertainty, route topology, repeat-scan change and replay.	pp. 3-15
S10	UGTS_KC_4_1_1_ATLAS_SPATIAL_SEED_SLAM_POCO_X7_PRO_MERGE_REPORT(1).pdf	Camera/IMU proposal path, bounded SLAM front end, stable IDs, ordered verifier and KSEED seed-plus-evidence boundary.	pp. 2-14
S11	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped identities and explicit release graphs instead of one misleading global ladder.	pp. 1-6
S12	UGTS_KC_Definition_Breaker_DB_0_1_Concept(1).pdf	Rules as physical objects, safe preview, hinges, winding, replay, seeded grammar and reason-coded commit.	pp. 2-24
S13	04_UGTOMS_RULEFORGE_1_0.pdf	Reusable rule-world host: typed actions, visibility, chance, conflict resolution, terminality and deterministic replay.	pp. 2-18
S14	06_UGTOMS_PROOFWEAVE_1_0.pdf	Proof graphs, small kernel, scoped certificates, trust boundaries and proposal-only search.	pp. 2-18

ID	Registered source	Use in this cartridge	Pages
S15	03_UGTOMS_FORGE_1_0.pdf	Parts, interfaces, datum frames, assembly order, inspection and infrastructure/fabrication boundaries.	pp. 2-16
S16	05_UGTOMS_VOLT_1_0.pdf	Typed networks, ports, capacities, events, faults, model adapters and physical-evidence separation.	pp. 2-17
S17	02_UGTOMS_SONORA_1_0.pdf	Executable notation, exact rational timing, cyclic structures, voices and downstream audiovisual projections.	pp. 2-15

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_CIVITAS:

D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)	
Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

CIVITAS cartridge package

Package record	Required role
city_profile.json	Authority mode, coordinate/time/unit profiles, governance, privacy and adapter capabilities.
place_graph.json	District, parcel, building, level, room, portal and stable asset identities.
mobility_graph.json	Route nodes/edges, movement modes, accessibility requirements, capacities and schedules.
utility_graph.json	Typed infrastructure nodes, edges, loads, reserves, isolation and dependencies.
policy_registry.json	Content-addressed rules, permits, stewardship, service obligations and exceptions.
ledger.jsonl	Observed proposals, accepted/rejected events, atomic patches, checkpoints and lineage.
scenario_branches/	Pure planning, game and counterfactual branches with explicit parent checkpoints.
certificates/	Route, impact, change, replay and publication certificates.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
                laws, exactness, effects, capabilities, dependencies,
                order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

Operator CVT041 constrained_route_query
domain: CityState x RouteRequest → RouteCertificate
profiles: mode, access, time, capacity, policy, uncertainty
exactness: exact graph traversal over declared finite graph; measurements remain interval-bound
effects: pure query only
authority: may propose a route-status patch only through the verifier

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal City Object and Authority Classes

C = (P, A, M, U, R, O, S, E, B, Delta, L, Cert)

P places and portals

A assets, actors and stewardship

M mobility and accessibility graph

U utilities, capacities and dependencies

R rules, permits, schedules and obligations

O observations with uncertainty and provenance

S simulated/planned scenario branches

E ordered accepted/rejected city events

B finite work, privacy and publication budgets

Delta atomic city patches

L checkpoints, lineage and replay

Cert route, impact, evidence and replay certificates

Authority class	Meaning	May mutate city state?
NORMATIVE	Current declared city/world model: stable places, routes, services, rules and schedules.	Only through verified patch.
OBSERVED	Sensor, scan, inspection, human note or external feed with source and uncertainty.	No - proposal only.
SIMULATED	Future plan, disaster drill, game branch, demand forecast or construction scenario.	No - branch-local only.
PRESENTATION	Map, render, dashboard, narration, haptic or printed view.	Never.
EXTERNAL AUTHORITY	A mature municipal, building, transit or utility system accessed through an adapter.	Only through negotiated import and verifier.

IDENTITY SPLIT

city object ID != coordinate != drawing != current condition != owner != observation != scenario copy. A doorway can be remeasured, blocked, repaired or displayed differently without becoming a new doorway.

Stable Place, Asset and Actor Identity

CIVITAS: one city, several typed layers

A road, room, utility line, rule and observation may share coordinates without sharing identity or authority.

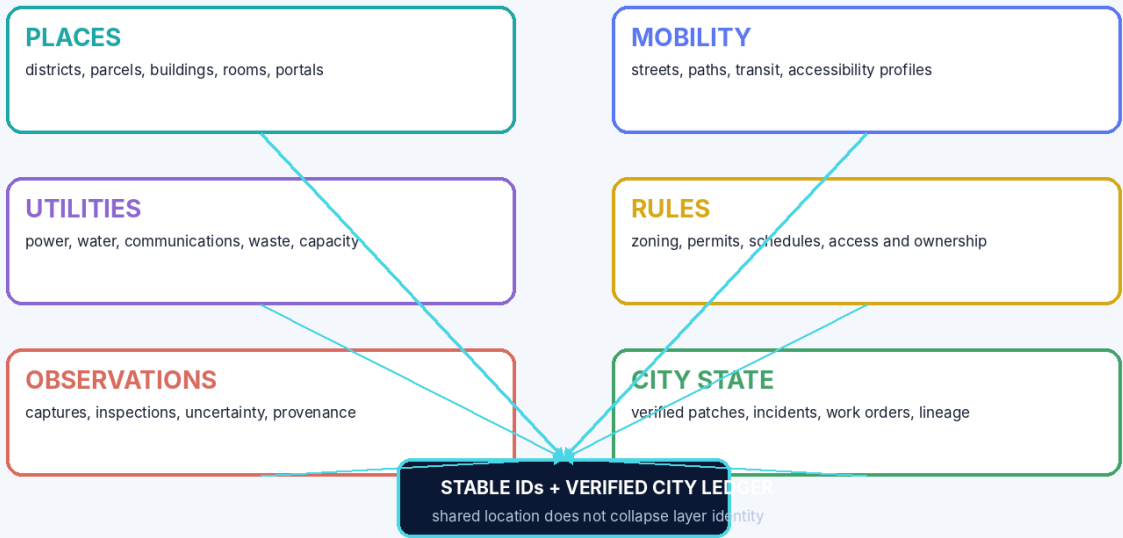


Figure 1. One city contains several typed layers whose shared coordinates do not collapse identity or authority.

Identity family	Required distinctions
Place hierarchy	city, district, parcel, building, level, room and subspace are stable nodes; containment and adjacency are separate relations.
Portals	door, crossing, lift, platform, gate and transfer point are topological edges with state, direction and mode support.
Assets	signs, trees, equipment, furniture, sensors and service components carry steward, revision, location and evidence.
Actors	resident, visitor, operator, maintainer, responder, simulated agent and anonymous demand class are typed roles.
Ownership/stewardship	owner, operator, maintainer, tenant, public-access steward and emergency authority are separate relations.
Revision lineage	renaming, relocation, subdivision, renovation and replacement are explicit transitions, not ID reuse.
TOPOLOGY RULE	
A portal connects declared places under a route profile. A visible opening, shared coordinate or scan gap does not create a route edge by itself.	

Mobility, Accessibility and Route Topology

```
RouteRequest = (origin, destination, time, mode, access_profile,
               capacity_policy, risk_policy, unknown_policy)
```

```
admit(edge) = supported_mode
              and schedule_open
              and passability in allowed_states
              and clearance_interval >= required_clearance
              and slope/effort interval within profile
              and capacity/reservation guard passes
              and evidence age + uncertainty satisfy policy
```

Route dimension	Examples	Boundary
Movement mode	walk, wheel, cycle, transit, service, stretcher, responder	Mode is a profile, not a value judgment about a person.
Physical access	clearance, turn radius, surface, slope, lift, curb, rest point	Use intervals and unknown states; do not infer from appearance alone.
Sensory/communication	audible/visual cues, tactile path, quiet route, sign language or assistance point	Availability and reliability are time- and evidence-bound.
Effort and pace	distance, incline, transfer count, waiting, energy model	No universal accessibility score is promoted.
Capacity and schedule	occupancy, reservations, service window, maintenance, festival closure	The same edge may be legal for one time/profile and blocked for another.
Unknown policy	reject, allow-with-warning, seek evidence or branch simulation	Unknown never silently becomes open or safe.

ROUTE CERTIFICATE

A route result returns reachable, unreachable or unknown plus the exact edge sequence, profiles, failed guards, interval margins, evidence ages, state hash and non-claims. It does not certify real-world safety by itself.

Utilities, Capacity and Infrastructure Dependencies

Network family	Typed elements	Typical verified event
Power	sources, feeders, switches, loads, storage, reserve and isolation	outage, transfer, overload guard, maintenance isolation, return to service
Water/thermal	sources, pumps, pipes, zones, valves, storage, demand and quality profile	isolation, pressure/capacity incident, maintenance, restoration
Communications	nodes, links, coverage/service profile, bandwidth and backup dependency	service degradation, failover, repair, scheduled maintenance
Waste/logistics	collection nodes, routes, capacities, transfer and service windows	route closure, missed service, overflow proposal, alternate assignment
Transport	lines, stops, platforms, vehicles, timetables, transfers and disruptions	delay, diversion, platform closure, accessible shuttle activation
Buildings/services	lift, alarm, ventilation, clinic, shelter, charging and public service dependencies	fault, inspection, capacity change, emergency profile activation

Impact propagation

```

impact(proposal, dependency_graph, profile, budget)
  -> directly affected nodes/edges
  -> guarded downstream candidates
  -> unresolved cycles / unknown adapters
  -> proposed patch set + certificate

```

No dependency edge, no automatic causal claim.

CYCLE POLICY

A circular dependency is rejected unless the profile declares initialization, update order, fixed-point/convergence semantics and a finite budget. Visual arrows cannot hide an unresolved loop.

Policies, Permits, Schedules and Stewardship

Rule object	Core fields	What it controls
City policy	stable ID, jurisdiction/profile, domain, effective dates, exceptions, source and version	route use, access, operation, construction, service or publication constraints
Use/zoning profile	allowed, conditional, restricted, prohibited or unknown classifications	fictional game worlds, planning branches or imported authoritative profiles
Permit/authorization	proposer, approving profile, scope, evidence, expiry and revocation	construction, closure, temporary service, event setup or data promotion
Schedule	clock/time-zone/calendar profile, interval, recurrence, priority and exceptions	opening, service, maintenance, festival, curfew and phased works
Stewardship relation	owner, operator, maintainer, tenant, public steward and emergency role	who may propose, approve, inspect, repair, publish or override under a bounded profile
Service obligation	target service, population/profile, time, capacity and evidence	monitoring and simulation goals without pretending legal enforceability

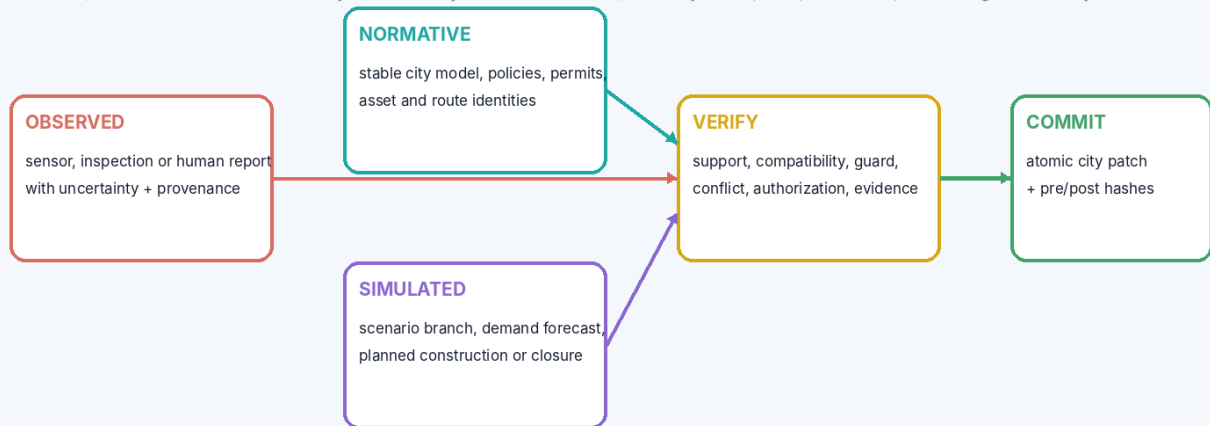
LEGAL / POLICY BOUNDARY

CIVITAS can store and evaluate a named policy profile. It does not invent current law, legal validity, ownership, emergency authority or permit status. Production use requires authoritative sources and governance.

Observation, Simulation and Normative Authority Split

CIVITAS authority split

Observed, simulated and normative city records may inform each other, but only an explicit promotion path changes authority.



A model prediction never silently becomes a real closure, permit, route fact or public warning.

Figure 2. Observed, simulated and normative city records inform one another but require explicit promotion to change authority.

Proposal source	Required metadata	Promotion boundary
Phone/camera/SLAM	capture profile, calibration/model hashes, pose/support, uncertainty and privacy	may propose stable association, measurement or route-state change; never direct commit
Human inspection	inspector/role profile, time, object ID, method, result, uncertainty and attachments	policy decides whether one report is enough or corroboration is required
Utility/transit feed	adapter/version, time base, source identity, service profile and confidence	external authority must be negotiated; stale or conflicting feeds reject or branch
Simulation	parent checkpoint, model/version, assumptions, seed, budget and branch ID	remains scenario-local until separately authorized and evidenced
Manual planning edit	author, rationale, affected IDs, permit/profile, dependency closure and preview	requires support, compatibility, authorization, conflict and invariant guards

Verified City-Event and Atomic-Patch Pipeline

capture / planner / game / operator / external feed -> typed city proposal -> local place/time/network support -> semantic, mode, jurisdiction and profile compatibility -> evidence, interval, schedule, capacity and authorization guards -> dependency closure + conflict classification -> verified proposal -> deterministic same-time resolution -> atomic city patch -> pre/post hashes + checkpoints + lineage -> optional map, alert, simulation, game or public view	
Event type	Atomic patch examples
Closure	set edge/place/service status, exceptions, start/end, reason, evidence and steward.
Construction	create/revise/remove place, portal, route, asset and dependencies under a branch or permit profile.
Maintenance	isolate, inspect, repair, test, restore and record unresolved findings.
Incident	blockage, outage, crowding, hazard or damage state with confidence and expiry/review.
Emergency profile	activate bounded rules, route permissions, service priorities and expiry with explicit authority.
Observation promotion	attach evidence, update an interval or state classification, and preserve the superseded record.
ONE COMMIT BOUNDARY	
A map editor, planner, sensor, AI model or game system may calculate anything it wants in its own proposal branch. The authoritative city changes only through one ordered verifier and atomic patch boundary.	

Deterministic Festival-Outage-Flood Demonstration

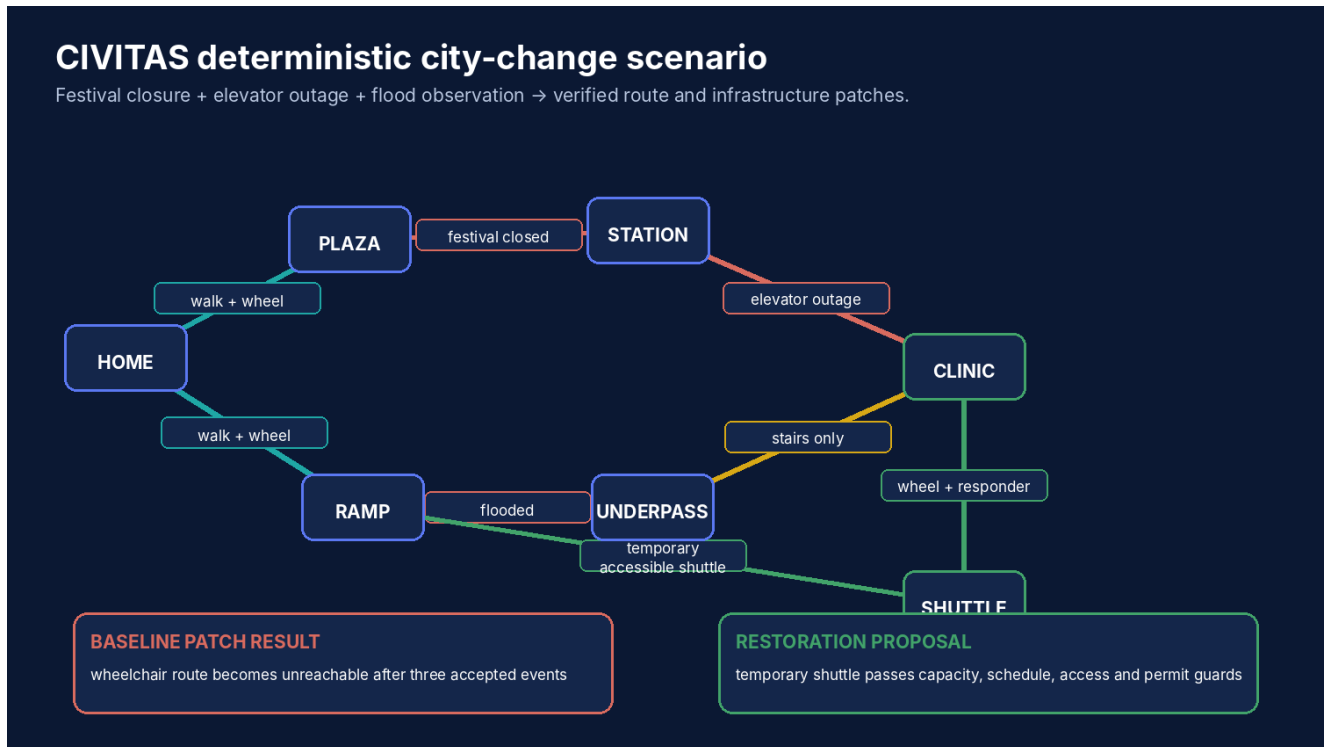


Figure 3. Three accepted disruptions remove the original accessible route; a verified temporary shuttle restores one.

Sequence	Proposal	Verifier result	Committed effect
1	Festival closes Plaza → Station edge from 07:00-23:00.	ACCEPT: schedule, permit and scope resolve.	Walking/wheeling edge closed for declared interval.
2	Clinic station elevator reports fault.	ACCEPT: stable lift ID, operator feed and maintenance profile pass.	Station → Clinic access becomes stairs-only.
3	Underpass flood observation.	ACCEPT: support, two reports and water-depth interval exceed guard.	Ramp → Underpass edge blocked for all public modes.
4	Wheelchair route query HOME → CLINIC.	UNREACHABLE: every path fails closure, lift or flood guard.	No city mutation; route certificate records exact failures.
5	Temporary accessible shuttle proposal.	ACCEPT: vehicle, stops, capacity, permit, schedule and operator authority pass.	Ramp → Shuttle → Clinic edges added with expiry.
6	Repeat route query.	REACHABLE under wheel profile with one transfer.	No mutation; new certificate references post-patch hash.

DEMO BOUNDARY

This is synthetic deterministic reference data. It does not prove an actual route is accessible, safe, legal or operational.

Queries, Packing, Privacy and External Adapters

Query	Returns
route(origin, destination, profile)	reachable/unreachable/unknown, path, margins, failed guards, evidence and state hash
impact(proposal)	direct and dependency-propagated candidates, unknown adapters, cycles and work budget
change(baseline, current)	stable-ID additions, removals, movement, state/measurement changes and within-tolerance items
why(object/event)	definition versions, observations, policies, dependencies, conflicts, commit and lineage
scenario_diff(branch_A, branch_B)	patches, route/service consequences, unknowns and first divergence
publication_view(policy)	redacted or aggregated city data with an internal hash-linked source record

Packing and reconstruction rules

- Use stable IDs, shared definitions, compact graph indices, Morton/prefix spatial keys, schedules and event deltas where exact reconstruction remains possible.
- A local code locates a record; it never replaces units, uncertainty, policy, topology, authority class or lineage.
- Store seeds only for declared deterministic scenario families. Real observations, external events and human decisions remain explicit novelty.
- Unknown codebooks, unresolved definitions, broken hashes or migration gaps reject rather than alias another city object.

Adapter family	Integration rule
GIS / mapping	Import geometry and identifiers through a versioned mapping; display layers do not become city authority automatically.
BIM / CAD / facility systems	Keep external model revision, coordinate transform, units and stable-object mapping explicit.
Transit / utility systems	Negotiate schemas, time bases, service states, capacities and authority scope before accepting events.
Simulation / game runtime	Operate on a branch or rule-world profile; commits stay inside declared authority.
Phone / scanner	Emit observations and association proposals with uncertainty; no unchecked map mutation.
Public app / report	Use privacy views, freshness labels, disclaimers and a separate internal authoritative ledger.

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

32/32 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique CVT IDs; every mechanism has family, name and contract.
Layer identity fixture	Place, route, utility, rule, observation and scenario records sharing coordinates remain distinct IDs.
Route fixture	Baseline accessible path exists; three accepted disruptions remove it; a permitted shuttle restores it.
Authority split	Observed and simulated records cannot mutate normative city state without a verified proposal.
Dependency fixture	Outage impact follows declared graph edges and stops at missing/unknown adapters.
Replay fixture	Accepted city patches reproduce the post-state hash and reject a tampered event.
Unknown policy	Insufficient clearance, stale evidence or ambiguous temporal intervals return UNKNOWN rather than OPEN.
Packing boundary	Stable keys and shared definitions round-trip without replacing uncertainty or lineage.

Explicit non-claims

- No deployed smart-city, municipal, building, utility, traffic or emergency operating system is claimed.
- No route, capacity, damage, evacuation, accessibility or infrastructure result is a real-world certification.
- No legal policy, permit, ownership, privacy or public-warning validity is established by the cartridge.
- No phone scan, model output, map picture or content hash may directly become city authority.
- No universal accessibility score, city ontology, causal model, compression ratio or performance result is claimed.
- No external system compatibility is claimed until an adapter, schema mapping and authority profile are tested.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete CIVITAS 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
CVT001	Identity	City profile	Declares fictional, planning, operational or observed-twin authority; coordinate, policy and time conventions are explicit.
CVT002	Identity	District identity	Stable district ID, boundaries, parent city and governance profile remain distinct from display labels.
CVT003	Identity	Parcel identity	Stable land/lot unit with geometry reference, use profile, revision and lineage.
CVT004	Identity	Building identity	Stable building ID, revision, parent parcel, levels, entrances and infrastructure dependencies.
CVT005	Identity	Level and room identity	Rooms and levels preserve identity through remapping, renovation and representation changes.
CVT006	Identity	Portal identity	Door, gate, crossing, lift or transfer point is a stable topological edge, not a hole in a picture.
CVT007	Identity	City asset registry	Road furniture, signs, trees, equipment and service assets use stable IDs and steward records.
CVT008	Identity	Actor and role profile	Residents, visitors, operators, responders and simulated agents have typed roles and authority limits.
CVT009	Spatial	Coordinate-frame profile	Handedness, axis, origin, units, scale and vertical datum are declared; transforms carry capability and error.
CVT010	Spatial	Nested place graph	City → district → parcel → building → level → room hierarchy plus explicit cross-links.
CVT011	Mobility	Route node	Stable waypoint, entrance, platform, room, crossing or transfer node with supported modes.
CVT012	Mobility	Route edge	Stable connection with length, direction, schedule, passability, capacity and evidence.
CVT013	Mobility	Mobility mode profile	Walking, wheeling, cycling, transit, service and responder modes are typed policy profiles.
CVT014	Mobility	Accessibility profile	Clearance, surface, slope, lift, rest, sensory and communication requirements are explicit query data.
CVT015	Mobility	Clearance interval	Closed interval with source, uncertainty and threshold policy; a point estimate cannot silently decide access.
CVT016	Mobility	Slope and effort interval	Grade, rise/run, effort or energy estimate under a declared model and uncertainty profile.
CVT017	Flow	Occupancy and capacity	Finite capacity, current load, reservation and safety/governance profile for spaces and edges.
CVT018	Flow	Schedule window	Opening, service, maintenance, event and curfew windows with calendar/time-zone profile.
CVT019	Transport	Transport line	Stable line/service identity with ordered stops, timetable profile, vehicle/capacity and disruptions.
CVT020	Transport	Stop and station	Stable access node with platforms, transfers, entrances, facilities and route evidence.
CVT021	Utility	Utility node	Power, water, communication, thermal or waste source/sink/storage node under a typed network profile.
CVT022	Utility	Utility edge	Typed directional or bidirectional connection with capacity, losses, isolation and maintenance state.
CVT023	Utility	Capacity and load state	Demand, supply, reserve and uncertainty with unit, time and model capability.

ID	Family	Mechanism	Core contract
CVT024	Utility	Infrastructure dependency graph	Explicit dependencies among buildings, transport, utilities and services; undeclared cycles reject.
CVT025	Rule	City policy definition	Content-addressed rule with domain, authority, jurisdiction, schedule, exceptions and provenance.
CVT026	Rule	Use and zoning profile	Allowed, conditional, restricted or unknown uses under a declared fictional/planning/legal profile.
CVT027	Rule	Permit and authorization	A proposed construction, closure or operational change names the approving profile and evidence.
CVT028	Rule	Ownership and stewardship	Owner, operator, maintainer and public-access roles remain separate typed relations.
CVT029	Rule	Service obligation	Declares target service, access, maintenance or response commitments without claiming legal enforceability.
CVT030	Observation	City observation	Sensor, inspection or human report with support, time, source, uncertainty, privacy and provenance.
CVT031	Observation	Uncertainty and provenance	Position, relation, quantity and classification confidence remain attached to the observation and adapter.
CVT032	Observation	Stable association proposal	Matches an observation to an existing asset/place under reason-coded identity and uncertainty guards.
CVT033	Event	Incident proposal	Proposes blockage, outage, crowding, hazard or damage state; no direct city mutation.
CVT034	Event	Closure event	Atomic edge/place/service closure with scope, start/end, mode exceptions, reason and lineage.
CVT035	Event	Construction event	Creates, removes or revises places, routes, assets and dependencies through a permitted patch.
CVT036	Event	Maintenance event	Schedules and commits inspection, isolation, repair, test and return-to-service transitions.
CVT037	Event	Emergency-mode event	Activates a bounded scenario/profile with explicit authority, expiry and non-claim boundaries.
CVT038	Authority	Same-time conflict policy	Deterministic priority, compatible merge or conflict event for simultaneous city proposals.
CVT039	Authority	Atomic city patch	Finite node, edge, rule, schedule, utility and evidence updates with pre/post state hashes.
CVT040	Authority	Checkpoint and replay	Snapshots plus ordered events reconstruct city state and stop at the first divergence.
CVT041	Query	Constrained route query	Returns reachable/unreachable/unknown route plus reason trace under mode, time, capacity and policy.
CVT042	Query	Dependency impact query	Propagates a proposed outage or construction patch through declared infrastructure dependencies.
CVT043	Query	Accessibility parity query	Compares routes and service outcomes across access profiles without reducing accessibility to one score.
CVT044	Query	Repeat-state change report	Stable-ID baseline/current comparison with movement, state, measurement and unknown classifications.
CVT045	Scenario	Scenario branch	A simulated festival, outage, construction or future plan remains a branch until explicitly promoted.
CVT046	Packing	Finite spatial and network keys	Hash-bound local codes, Morton/prefix keys and graph indices locate records but never replace complete state.
CVT047	Adapter	External city-system adapter	Versioned GIS, building, transit, utility or simulation oracle behind typed import/export and capability ports.
CVT048	Governance	Publication and privacy view	Redacts or aggregates sensitive fields while preserving a hash-linked authoritative internal ledger.

Claims Ledger

Claim	Disposition	Reason
CIVITAS is a complete city operating system.	REJECT	It is a cartridge contract and reference model, not a deployed municipal platform.
A route query proves real-world safety.	REJECT	Synthetic or stale data cannot certify evacuation, accessibility, traffic or emergency safety.
A 3D scan or image may directly rewrite the city model.	REJECT	Observations remain proposals with uncertainty, stable association and promotion gates.
Same coordinates imply the same room, route or utility.	REJECT	Stable IDs, layer, sheet, role and lineage remain part of identity.
A simulated plan is an observed fact.	REJECT	Normative, observed and simulated authority classes are separate.
Content hashes prove legal ownership or permit validity.	REJECT	Hashes prove unchanged content only.
Accessibility can be reduced to one universal score.	REJECT	Mode, person, time, route, effort, communication and policy profiles remain explicit.
Network capacity models are universally accurate.	REJECT	Every flow/utility result is model-, data- and time-profile bound.
A local code or Morton key contains the whole city.	REJECT	Keys locate finite records and must remain bound to definitions, uncertainty and lineage.
CIVITAS replaces mature city, building or utility tools.	REJECT	Use versioned external adapters when conventional systems are safer or more capable.
Emergency mode may bypass authorization and lineage.	REJECT	Emergency profiles remain explicit, bounded, reason-coded and replayable.
The cartridge is independently validated for public deployment.	REJECT	Only deterministic specification self-checks and synthetic fixtures are supplied.

High-Impact Application Profiles

Profile	What the cartridge enables
Living City Sandbox	Build a SimCity-scale world where zoning, transit, utilities, accessibility and incidents are literal rule objects instead of hidden engine code.
Infrastructure Heist	Plan a stealth/tactics game around power, doors, transit, schedules, crowds and service dependencies with replayable world changes.
Accessibility-First Festival Planner	Preview closures, crowding, lifts, quiet routes and temporary services under multiple access profiles before publishing a plan.
Disaster Twin Replay Lab	Fork a city checkpoint, inject bounded outages/floods/closures, compare routes and replay exactly why a branch failed.
Neighborhood Builder Kit	Let children and communities construct streets, rooms, services and rules with physical tiles or Definition Breaker objects.
Pocket Civic Atlas	Merge phone observations into an offline, uncertainty-aware ledger without pretending a camera automatically knows the city.

Final Formal Definition

FORMAL DEFINITION

UGTOMS CIVITAS 1.0 is a component-scoped city-world cartridge in which places, portals, assets, actors, mobility modes, accessibility profiles, utility networks, capacities, schedules, policies, observations, scenario branches, conflicts and city changes are typed, content-addressed and replayable substrate data; observed and simulated inputs remain proposals; and only a verified atomic city patch may alter authoritative city state and lineage.

Implementation sequence

- Freeze schemas for CityProfile, PlaceGraph, MobilityGraph, UtilityGraph, CityPolicy, CityProposal and CityCertificate.
- Implement the exact finite graph, schedule, interval, dependency and replay oracle with the synthetic festival scenario.
- Add a RULEFORGE living-city sandbox and a browser visualizer whose map remains downstream of the ledger.
- Build read-only GIS/BIM/transit adapters with explicit identity mapping, units, time and authority negotiation.
- Promote any real operational use only after domain experts, authoritative data sources, accessibility co-design, privacy governance and field validation.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.